Sahaj Bhadja

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Summary

Junior software engineer with 2+ years building **C++/Python systems**, **Linux-based compute workflows**, and **LLM-assisted development pipelines** for reliable, high-throughput scientific and data-intensive applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **Python-based scientific data services** and interfaces for 500+ users, reducing query latency from **1.2s to 350ms** while improving correctness checks in analytics workflows.
- Integrated **AI-assisted debugging and documentation** into release cycles for 5K+ monthly jobs, improving incident triage speed and maintainability across cross-functional engineering teams.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **production APIs and UI modules** using React/Flask for 10K+ daily requests, translating product needs into tested, maintainable features in Agile sprint delivery.
- Improved **CI/CD and automated test workflows** via script-driven validation and defect triage, reducing release turnaround by **60%** with sustained deployment quality.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **compute-intensive experimentation pipelines** over 100K+ trajectory logs, enabling reproducible benchmarking and high-confidence evaluation for safety-sensitive model behaviors.
- Prototyped **LLM-assisted analysis tooling** to summarize experiment deltas and generate implementation-ready technical notes, improving handoff efficiency between research and engineering.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **scalable backend components** with Python/Java, AWS Glue, and S3, improving response time by **30%** and supporting reliable distributed processing pipelines.
- Maintained **20+ CI-integrated test suites** using Jenkins, PyTest, and Mockito on Linux/Windows environments, increasing pipeline throughput by **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Implemented **MERN + WebSocket services** for eight workflows and 2M+ synchronized records, improving consistency and monitoring for high-frequency event-driven operations.
- Automated **containerized build and deployment** with Docker/Jenkins, cutting rollout time by **50%** through repeatable scriptable release processes and health checks.

Skills

Languages: **C++**, **Python**, Java, JavaScript/TypeScript, SQL, Bash
Scientific/Systems: Linux/UNIX, multithreaded workflows, performance profiling, reproducible benchmarking
LLM Tooling: **ChatGPT/Claude/Copilot workflows**, prompt-driven code generation, API-based automation
Build & Delivery: CMake/Make, Docker, Jenkins, GitHub Actions, CI/CD, test automation (PyTest, Mockito)
Engineering: Git/GitHub, Agile/Scrum, code reviews, technical documentation, cross-functional collaboration

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, and MLFQ engines** with metrics and visualization, reducing debugging effort by **20%** through deterministic, testable simulation outputs.

LLM Code Review Assistant | *Python, OpenAI API, FastAPI, PostgreSQL*

- Built a **prompt-orchestrated review pipeline** that analyzes pull requests, flags risk patterns, and drafts actionable summaries, improving engineering feedback loops and triage quality.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized REST processing service** with AWS deployment automation and validation checks, improving reliability and operational continuity for production data workloads.